E-Commerce Website

Full Stack Technologies (FST) Project Report

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3. DESIGN

3.1 Workflow Diagram and Explanation

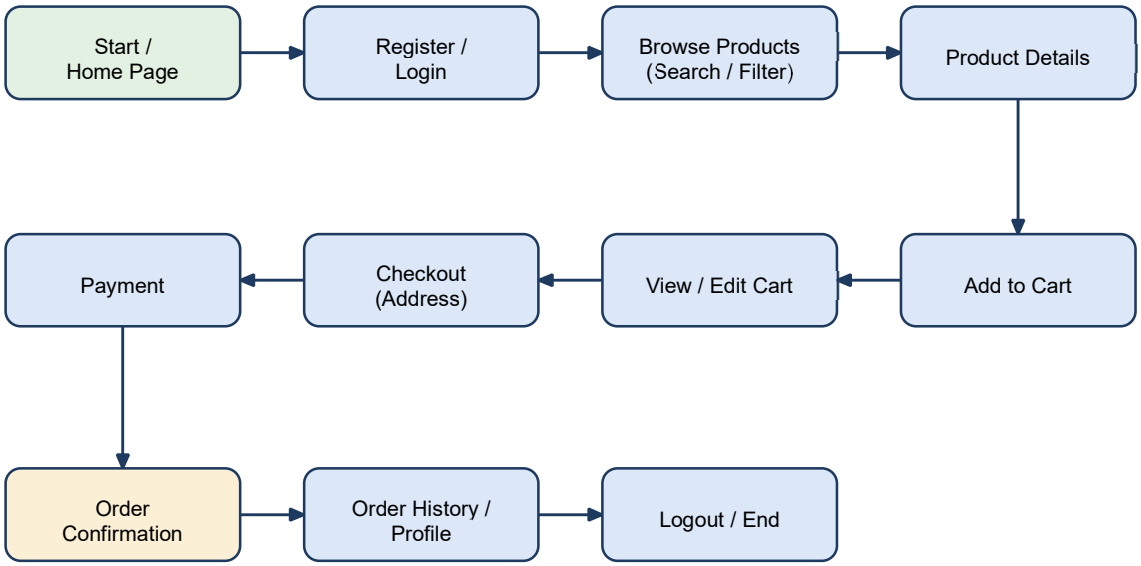
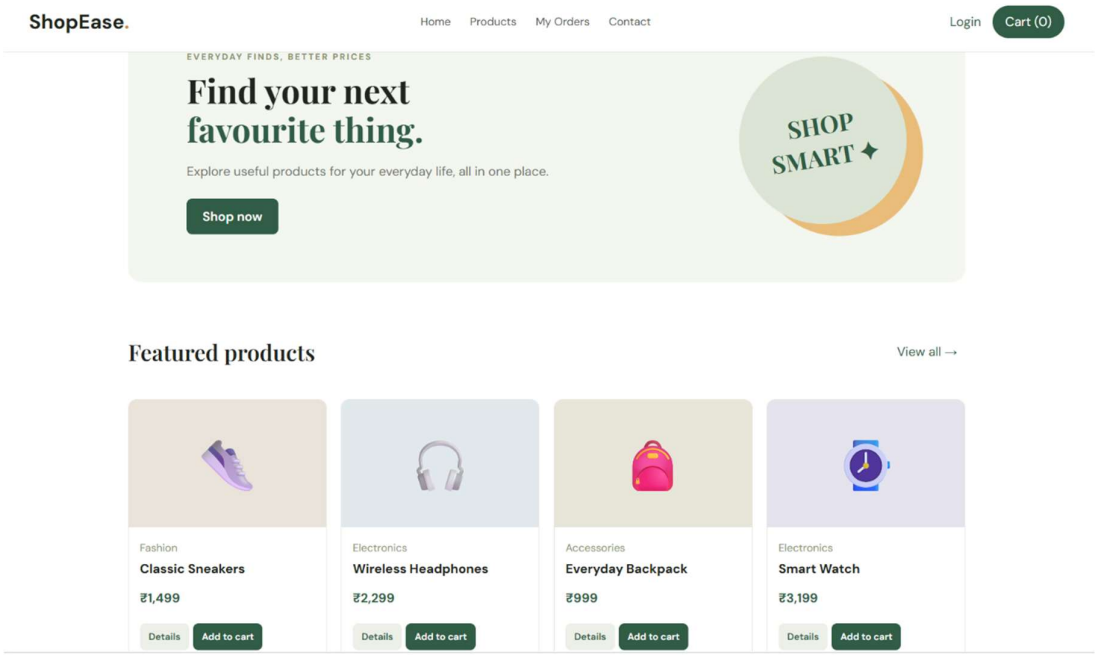


Fig 3.1: Workflow of the E-Commerce website

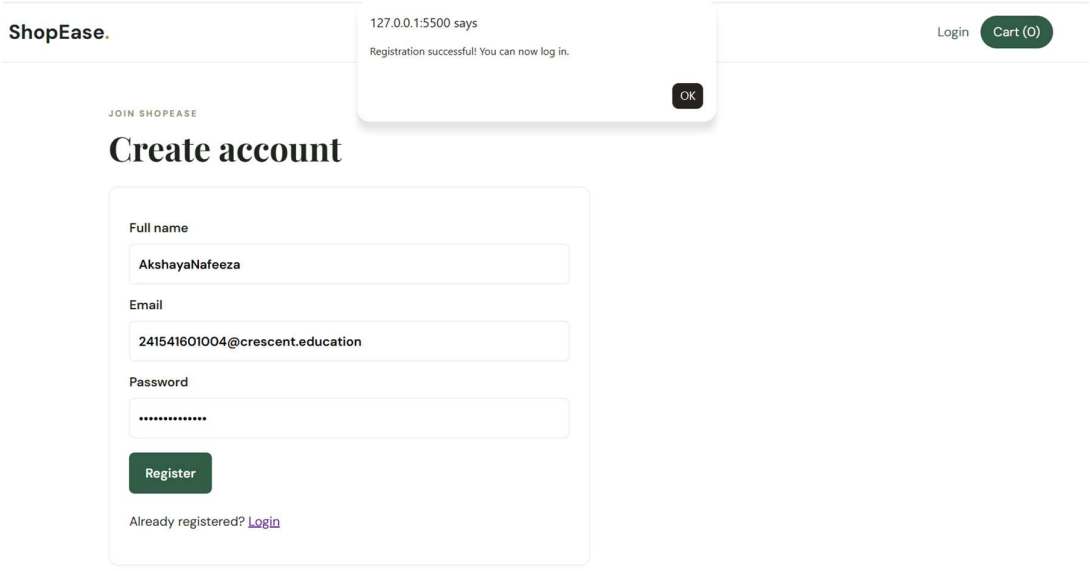
- **Home Page:** The user lands on the home page showing featured products, categories and offers.
- **Register / Login:** New users create an account; existing users log in. Details are validated before access is given.
- **Browse Products:** Products are listed by category, with search and filter options (price, category).
- **Product Details:** Shows image, price, description and the Add to Cart button.
- **Cart:** Selected items are stored in the cart. The user can change quantity, remove items and see the total.
- **Checkout and Payment:** The user enters the delivery address and selects a payment method.

- **Order Confirmation:** A success page with the order summary is shown and the order is saved in order history.

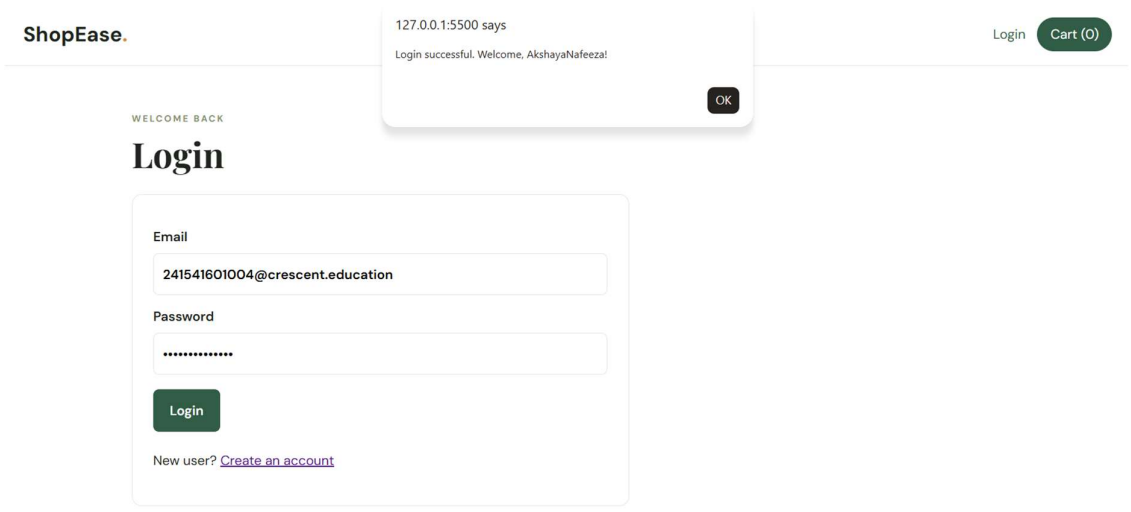
3.2 Web Page Screenshorts



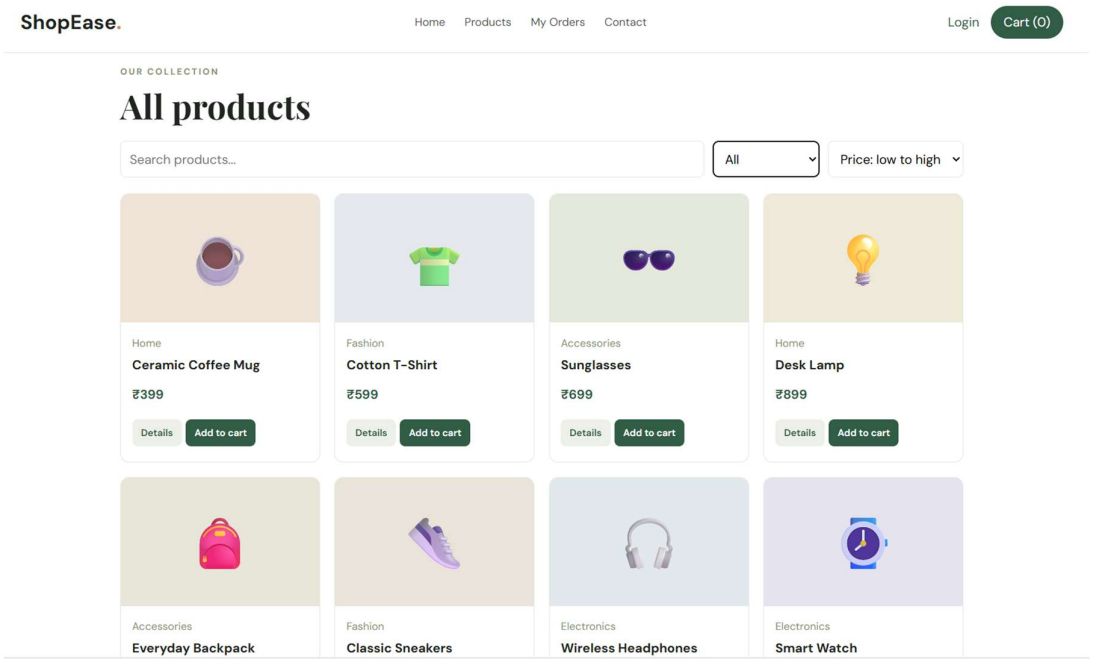
Home page



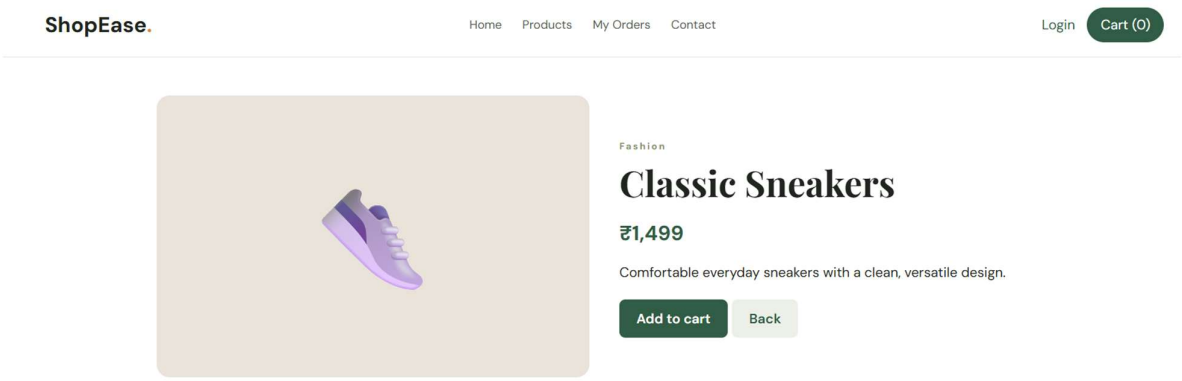
Registration page



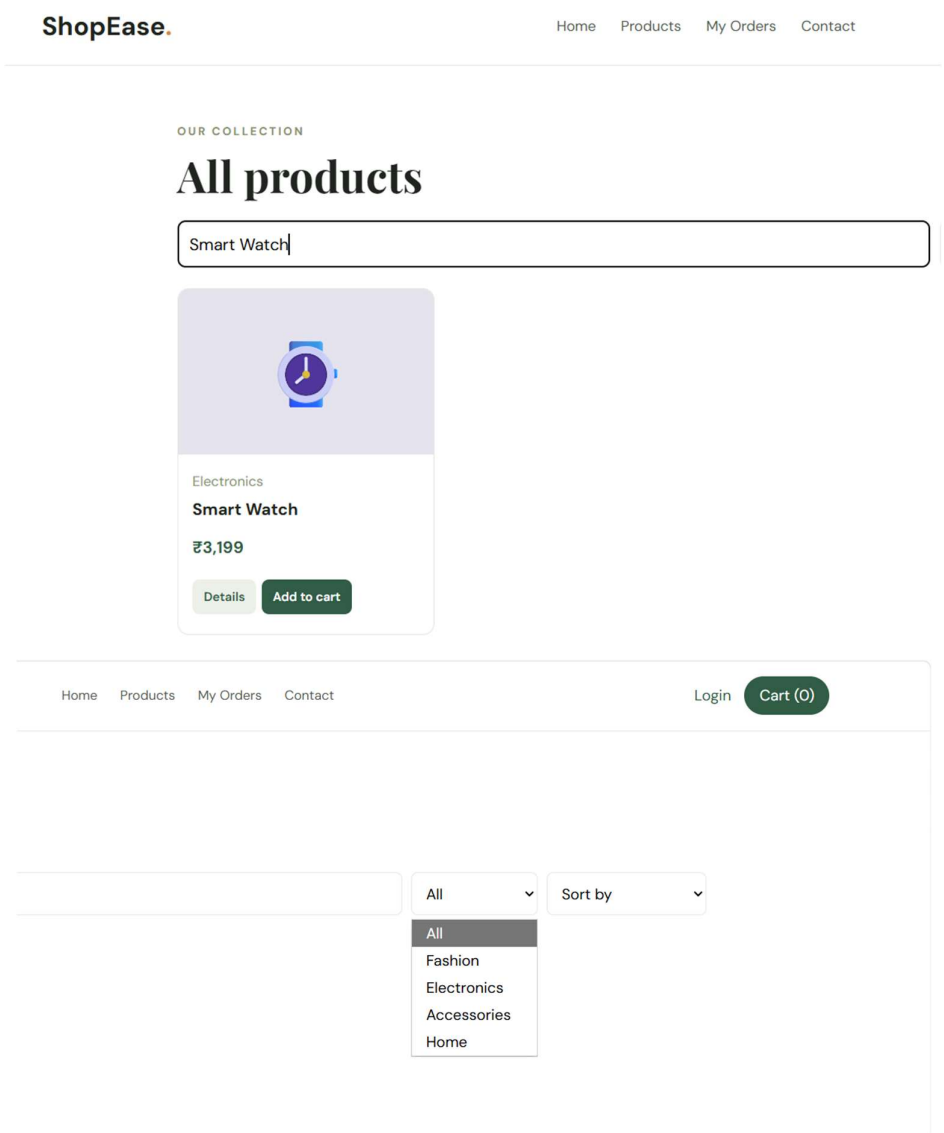
Login Page



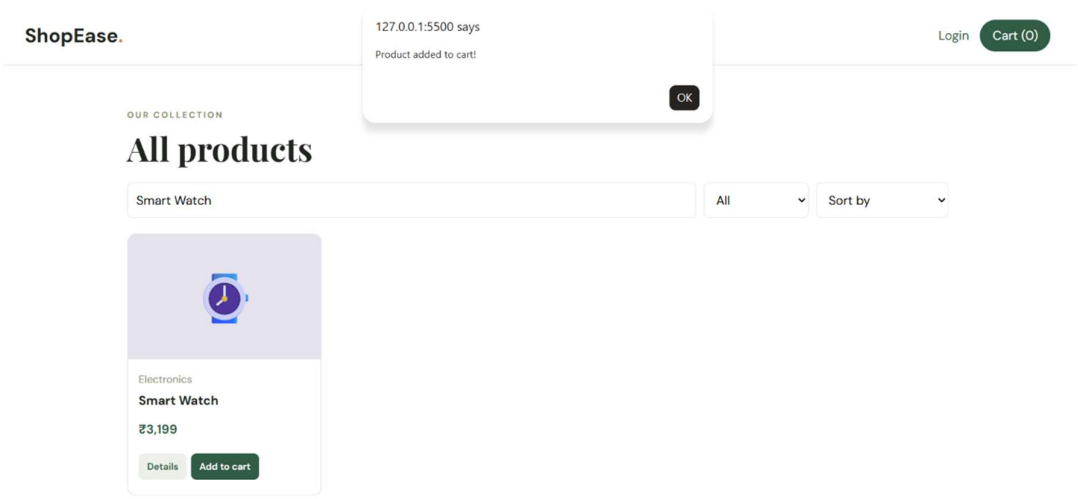
Product Listing Page



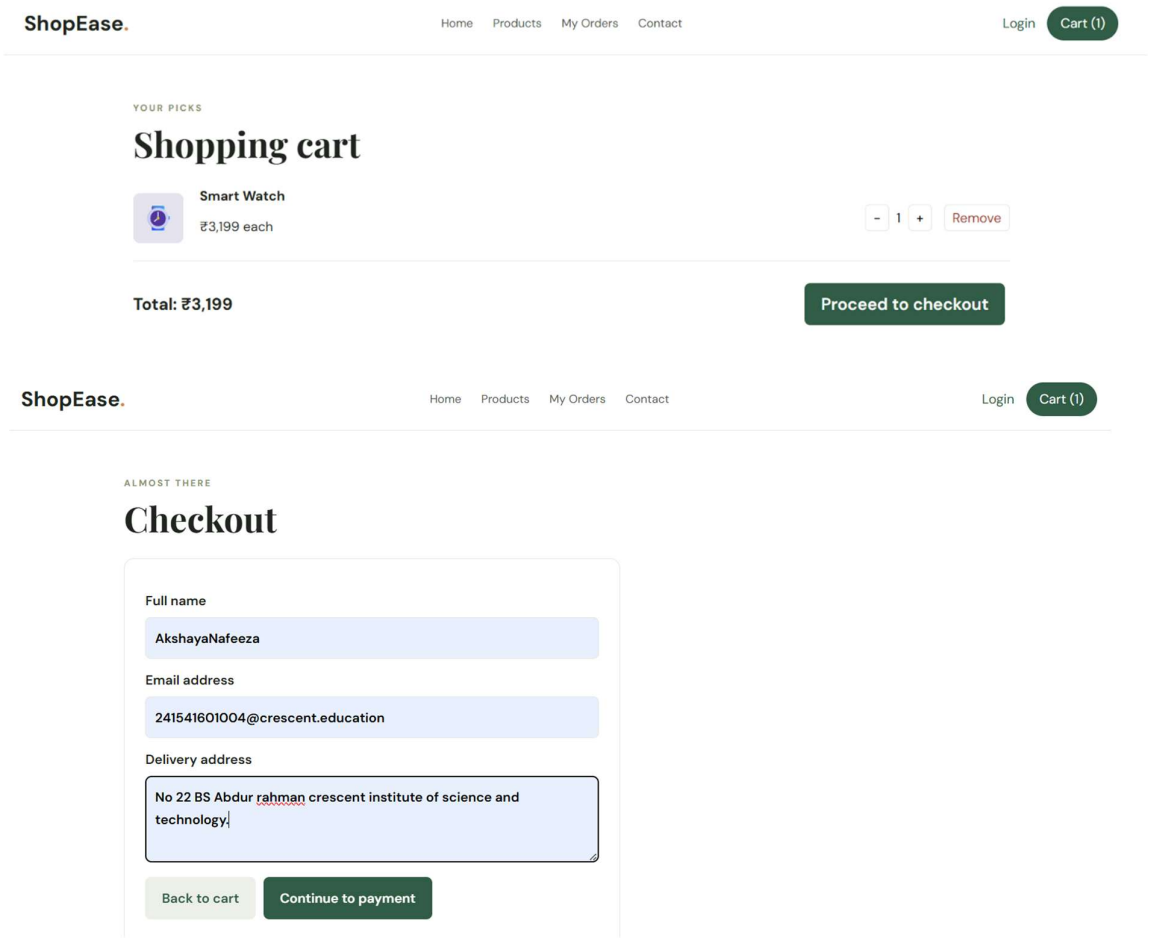
Product Details Page



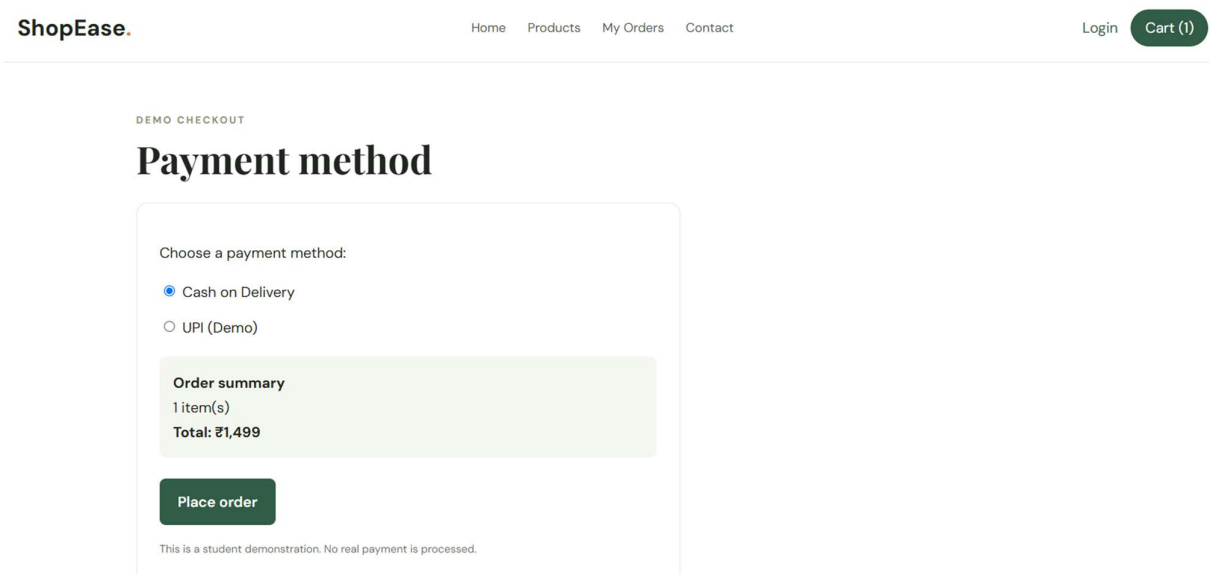
Search/ Filter results



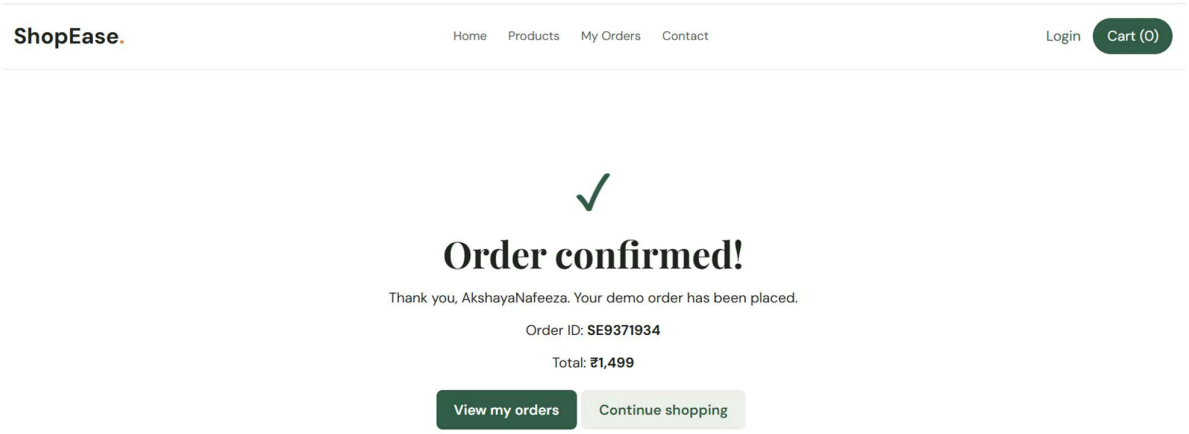
Shopping Cart Page



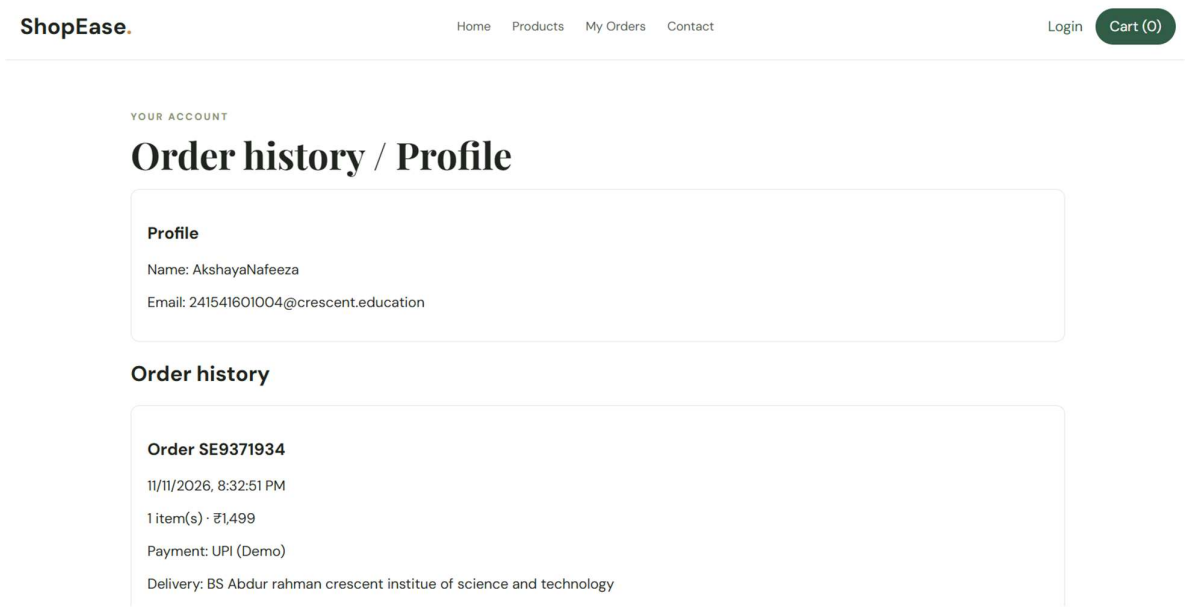
Checkout Page



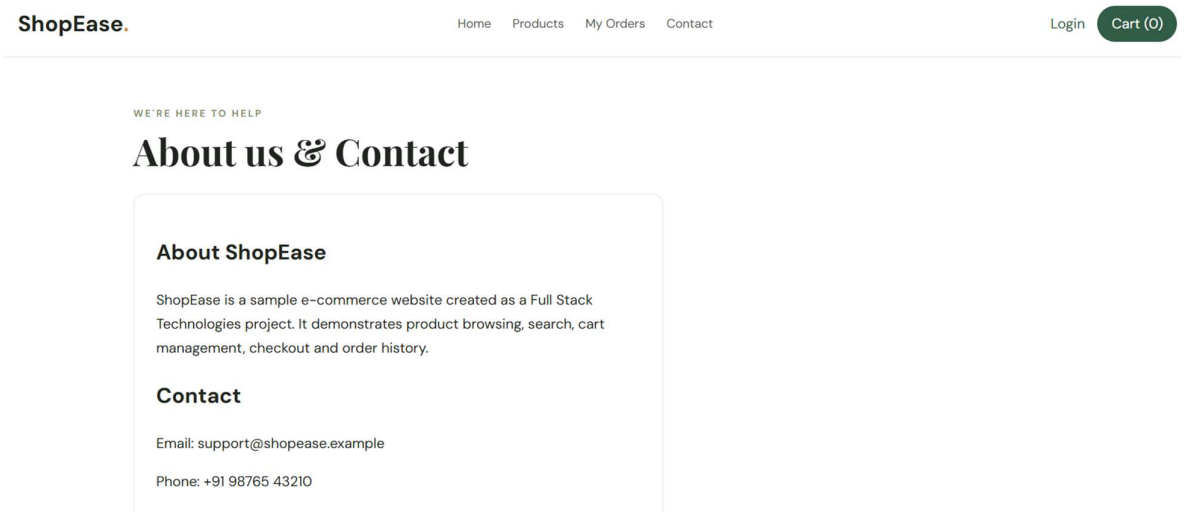
Payment Page



Order Confirmation Page



Order History/ Profile Page



Contact / About Us page

4. IMPORTANT CODE AND EXPLANATION

The sample code below shows the key modules of the website (HTML, CSS and JavaScript). Replace or adjust it to match your own project files.

4.1 Product Card Layout (HTML)

```
<div class="product-card">
  <img src="images/shoe.jpg" alt="Running Shoe">
  <h3>Running Shoe</h3>
  <p class="price">Rs. 1,499</p>
  <button onclick="addToCart(1)">Add to Cart</button>
</div>
```

Explanation: Each product is shown as a card with image, name, price and button. The button calls `addToCart()` with the product ID.

4.2 Responsive Grid (CSS)

```
.product-grid {   display: grid;   grid-template-columns:
repeat(auto-fill, minmax(220px, 1fr));   gap: 20px; }
@media (max-width: 600px) {
  .product-grid { grid-template-columns: 1fr; }
}
```

Explanation: CSS Grid arranges the cards automatically; the media query shows one column on mobile screens, making the site responsive.

4.3 Add to Cart using localStorage (JavaScript)

```
function addToCart(id) {   let cart =
JSON.parse(localStorage.getItem("cart")) || [];   const
item = cart.find(p => p.id === id);   if (item) item.qty
+= 1;   else cart.push({ id: id, qty: 1 });
localStorage.setItem("cart", JSON.stringify(cart));
updateCartCount(); }
```

Explanation: The cart is read from localStorage. If the product exists, its quantity increases; otherwise it is added. The updated cart is saved back, so it remains after page refresh.

4.4 Cart Total Calculation

```
function getTotal(products) {   const cart =
JSON.parse(localStorage.getItem("cart")) || [];   return
cart.reduce((sum, c) => {     const p = products.find(x =>
x.id === c.id);     return sum + p.price * c.qty;   }, 0);
}
```

Explanation: `reduce()` multiplies each item price by its quantity and adds them to give the cart total.

4.5 Login Form Validation

```
document.getElementById("loginForm").addEventListener("submit", e => {
  e.preventDefault();   const email =
document.getElementById("email").value.trim();   const pass  =
document.getElementById("password").value;
  if (!/^\S+@\S+\.\S+$/.test(email) || pass.length < 6) {
alert("Enter a valid email and a password of 6+ characters");
return;   }   window.location.href = "index.html"; });
```

Explanation: The form is stopped from submitting until the email format and password length are checked. Valid users are redirected to the home page.

5. HOSTING DETAILS

The website was hosted using **GitHub Pages**, which gives free static hosting directly from a GitHub repository.

Hosting Platform	GitHub Pages
Repository Name	_____
Live URL	https://<username>.github.io/<repo-name>/
Repository URL	https://github.com/<username>/<repo-name>

Steps followed

1. Create a new repository on GitHub (public).
2. Upload the project files (index.html, CSS, JS, images) or push using Git: `git init`, `git add .`, `git commit -m "first commit"`, `git push origin main`.
3. Open **Settings > Pages**, choose the **main** branch and **/ (root)** folder, then click Save.
4. Wait 1-2 minutes; GitHub shows the live URL of the website.
5. Open the URL and test all pages after hosting.

Screenshots after hosting

Paste screenshot here

GitHub repository / Pages settings

Fig 5.1: GitHub repository / Pages settings

Paste screenshot here

Live website opened from the hosted URL

Fig 5.2: Live website opened from the hosted URL

Paste screenshot here

Hosted website on mobile / another page

Fig 5.3: Hosted website on mobile / another page

6.CHALLENGES FACED

No.	Challenge	How it was solved
1	Responsive design	Layout broke on small screens. Solved with CSS Grid, Flexbox and media queries.
2	Cart data lost on refresh	Cart items disappeared after reloading. Solved by storing the cart in localStorage.
3	Form validation	Invalid emails and empty fields were being accepted. Solved with JavaScript validation and regular expressions.
4	Image and file paths after hosting	Images did not load on GitHub Pages because of wrong paths and case-sensitive file names. Fixed with relative paths and consistent naming.
5	Cross-browser differences	Some styles looked different in different browsers. Tested in Chrome, Edge and Firefox and adjusted the CSS.
6	Git and deployment errors	Faced push and branch errors at first. Learned the basic Git workflow (add, commit, push) to resolve them.

7.CONCLUSION

The E-Commerce website was designed and developed to let users browse products, add them to a cart, and complete the checkout process through a simple, responsive interface. The project gave practical experience with HTML, CSS and JavaScript, form validation, client-side data storage, version control with Git, and deployment on GitHub Pages.

Future enhancements: a back-end with a database (Node.js / MongoDB), secure authentication, a real payment gateway, an admin dashboard, and product reviews and ratings.